THE NOVA RESEARCH CURRICULUM

Computer Science and skills

Grade 8 · Term 1

Student edition

Outcomes: CS-8.1 a small Python project with testing and debugging; RO-8.1; FL-8.1; RP-8.1; AR; PE.

Lesson notes

- **CS-8.1 (Weeks 1–12):** choose a project: a quiz, a calculator or a number-guessing game. Plan: what it must do, a flowchart, the functions needed. Build with variables, `if`/`elif`/`else`, loops, lists and functions; use `random` for the game. Test with a checklist of normal, edge and wrong inputs. Debugging: syntax errors (typing), runtime errors (crashes, e.g. dividing by zero), logic errors (wrong answer). Use `try`/`except` for bad input.
- **RO-8.1:** the engineering design cycle: find a local problem (for example, plants dying at school), research, ideas and sketches, a first prototype.
- **FL-8.1:** business plan part 1: idea and customers; fixed and variable costs; pricing.
- **RP-8.1:** independent project: a clear question; find and record sources; choose a method (survey, experiment or interviews with permission).
- **AR, PE:** observational drawing and Islamic calligraphy; basketball; a personal fitness plan; health talk on first aid and CPR awareness.

Workbook pages

Page	Task
1	Project plan and flowchart.
2	Test checklist and bug log.
3	Design-cycle log: problem and ideas.
4	Cost and pricing sheet.
5	Research question and source log.

Question bank

#	Outcome	Question	Your answer
1	CS-8.1	A missing colon after `if x > 5` causes a ... error.	logic / syntax / runtime / network
2	CS-8.1	A program that runs but gives the wrong answer has a ... error.	syntax / logic / spelling / hardware
3	CS-8.1	`try` and `except` help a program handle bad input without crashing. True or false.	True / False
4	CS-8.1	Which module gives random numbers?	math / time / random / os

#	Outcome	Question	Your answer
5	CS-8.1	Name three kinds of test input.	_____
6	RO-8.1	The first step of the design cycle is to …	build / test / identify the problem / sell
7	FL-8.1	Rent for a shop is a … cost.	variable / fixed / hidden / zero
8	RP-8.1	Interviews need the person's permission. True or false.	True / False